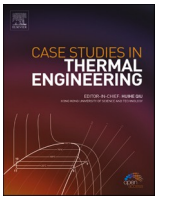
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Enhancement of heat transfer using ultrasonic waves: Experimental investigation and empirical correlations

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ABSTRACT

This work aimed to investigate the effect of ultrasonic waves on heat transfer enhancement under a subcooled pool condition. For this purpose, a cylindrical rod is submerged in distilled water and a 1.7 MHz ultrasonic transducer was utilized to generate ultrasonic waves. Three main parameters were examined including loaded heat flux, water temperature, and heights (distance from ultrasonic transducer). A Particle Image Velocimetry (PIV) is used as a non-intrusive flow visualization system to visualize how ultrasonic waves propagate in a water medium. The results obtained without imposing ultrasonic waves are compared with that of the results where ultrasonic waves are imposed, the average heat transfer rate with imposing ultrasonic waves is increased by 133%, while the increasing in enhancement factor is about 2.4 times at (H = 22 mm). Meanwhile, the increase in water temperature led to decreases in the heat transfer rate, this decreasing in the heat transfer rate is the most with Tw = 50 °C and it's around 46.8%, while it's about 17.2% with Tw = 40 °C under similar conditions. Effect of heated surface height is also investigated and analyzed, the ultrasonic waves are influential in the case of changing height and improved the enhancement factor by 3.52 times at (H = 65 mm). By using the database obtained under test conditions, empirical correlations were proposed to evaluate the real benefits of using ultrasonic waves in heat transfer enhancement.

1. Introduction

Enhancement of heat transfer in power facilities and energy systems is an important factor for the efficiency of these systems, safety and economic issues. As an example, spent fuel pool in a nuclear power station, high-temperature spent fuel is produced during the maintenance of nuclear power reactors. Firstly, the fuel is held in an on-site storage facility (at reactor) in spent fuel pool under atmospheric pressure for extremely long periods of time [1–3], then they stored in dry and air-cooled casks before they are reprocessing or refabricating depending on the country's authorized nuclear waste management policy [4]. In spent fuel pool, as the radioactive fission product isotopes in the fuel decay, they release a considerable quantity of thermal energy and reaches high temperatures. Therefore, the spent fuel requires to be efficiently cooled to allow the temperature levels to decay and so avoid their damage. The use of heat transfer enhancement techniques to enhance the cooling performance can be useful in this case for ensuring the fuel safety and integrity [5]. Using ultrasonic waves is one of the most promising techniques to enhance the cooling performance without modifying the original system or adding any additional structures. For decades, ultrasonic waves have been presented as a proper technique to enhance cooling performance and increase critical heat flux [6–8]. Propagation of ultrasonic waves in a liquid medium causes several

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mechanical mechanisms, the two major mechanisms are acoustic cavitation and acoustic streaming [9–11]. Fand [12] proposed that the powerful mechanism in ultrasonic wave was the acoustic cavitation produced in the liquid and observed an increasing in cooling performance by 18%. Liu et al. [13] conducted an experiment to investigate the possibility of heat transfer enhancement by ultrasonic vibration using a commercial stainless-steel heater as a heating element exposed to ultrasonic waves with a frequency of 40 kHz. They found 284% increasing in cooling enhancement ratio in water condition at 30 °C. Masoud et al. [14] claimed through experiments that the ultrasonic transducer position can play an important role in cooling performance. They found that the greatest enhancement occurs when single ultrasonic transducer was used. Suzuki et al. [15] investigated experimentally the effect of water and aqueous solutions of ethanol on cooling performance and microbubble emission boiling with ultrasonic vibration under range of pool liquid condition. They informed that the ultrasound waves increase heat flux and accelerate the microbubble emission boiling, but the effect was not obvious for the 100 wt% ethanol. Kiani et al. [16] further investigated the heat transfer phenomenon at the surface of a stationary sphere exposed to ultrasonic waves with a frequency of 25 kHz. They detected a 1.02 to 4 times increasing in cooling performance, this increasing was the most at closer distance to the transducer surface. Tajik et al. [17] confirmed through experiment that the acoustic streaming generated by ultrasonic vibrations can enhance heat transfer performance up to 390%. They also showed that the increase in the ultrasonic power and the decrease in distance between the heating element and ultrasonic surface produce the higher heat transfer enhancement. Hyun et al. [18] investigated the effect of acoustic streaming generated by ultrasonic vibrations with a frequency of 28 kHz on cooling performance of heated plate. The results confirmed a 40C decreasing in plate temperatures and the liquid velocity near the heated plate is in a range between 10–100 cm/s. In addition, theoretical study based on a lump-energy method was performed to support the result in this investigation. All of the previous investigations described above have been focused on the heat transfer from a commercial stainless-steel heater, flat plate, or a very small hotwire with low frequencies. The objective of the present investigation is then to emulate the actually spent fuel assembly and performed experiments to obtain a systematic database of cooling performance under ultrasonic waves with a high frequency of 1.7 MHz. The effect of the ultrasonic waves, height, loaded heat flux and water temperature on heat transfer characteristics are examined. To better understanding the ultrasonic waves mechanisms and explain the reason for heat transfer enhancement, streaming velocity and its transient features are predicted using Particle Image Velocimetry (PIV) system. This investigation will pose a novel development of ultrasonic waves as a highly efficient cooling technique for spent fuel assembly used in a Pressurized Water Reactor (PWR).

2. Experimental work

2.1. Experimental setup description

The schematic description for examining the effect of ultrasonic waves on the is shown in Fig. 1(a). A cylindrical rod with a diameter of 25 mm and a height of 90 mm is designed and manufactured using copper metal type C100. A real view of the cylindrical rod is shown in Fig. 1(c). With an A.C. stainless steel electric heater, the heat energy is generated inside the rod by Joule effect. The A.C. heater is inserted inside the rod in a wall-attached position, and the high thermal conductivity paste is applied between inside wall and the heater to improve the thermal conductivity and avoid any thermal resistance. To regulate the input power and therefore achieve the required heat flux and wall temperature, an AC voltage regulator was utilized, and the input power accuracy was evaluated about ±0.19%. Two thermocouples (type-K) were inserted in drilling holes created from the top of the rod, exactly in front of each other as shown in Fig. 2(b), to measure the average surface temperature. All of the created holes were filled with a sealant waterproof paste to prevent water penetration that may cause a major disruption in surface temperature measurements. The top side of the rod was

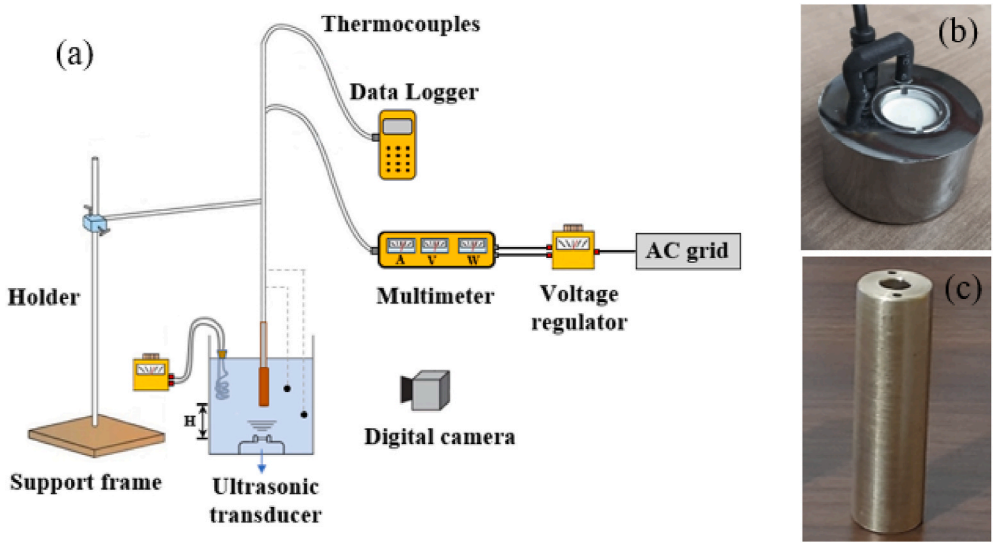


Fig. 1. (a) Schematic representation of the experimental setup; (b) ultrasonic transducer (c) cylindrical rod.

supported by hollow textolite rod as an insulation material with a diameter of 25 mm and a height of 30 mm. The heat exchange vessel with a size of 290 × 290 × 310mm filled with 200 mm-deep distilled water is utilized as cooling pool. The vessel is made of transparent Plexiglas sheet with 2 mm thickness to allow direct viewing the heat transfer phenomena at surface of the rod and propagation of ultrasonic waves in water. An ultrasonic transducer with a similar structure to that used in Ref. [19], frequency of 1.7 MHz, and ultrasound power of 14 W is utilized to generate ultrasonic waves. The ultrasonic transducer having a diameter of 35 mm, and a height of 25 mm and attached in the middle of the bottom side of the vessel, a real view of the ultrasonic transducer is shown in Fig. 1(b). The data acquisition system mainly includes eight thermocouples (type-K) with accuracy ±0.5% connected to digital data logger type (BTM-4208SD) and then the data saved to computer. The voltage and current at both ends of the heater are measured with a digital multimeter type (DT-830D). For recording the heat transfer phenomena on surface of the rod, a high-speed digital camera is employed, with the shooting done in transmitted light from a lamp.

2.2. PIV system

In order to better understanding how ultrasonic waves propagates in water medium and how they influence the rate of heat transfer, a Particle Image Velocimetry (PIV) system is used as a non-intrusive flow visualization system. In recent times PIV system is strongly recommended as an efficient approach for obtaining detailed velocity information and predicting flow patterns in a variety of scientific fields including fluid flow [20–22]. The PIV system used in present investigation consists of six basic components, which are as follows: (1) Pulsed laser to illuminate the water medium, (2) CCD camera as recording hardware having a resolution of 1600 × 1186 pixels, (3) PIV software and processor, (4) Control unit for synchronization of devices, (5) Tracer particles (seeding), and (6) Computer to obtain the velocity information from the tracer micro-particle positions. The PIV system works on the basis of capturing a pair of images of the water medium under investigation with a defined time interval between them. Each image is created by suspending the water medium with micro-particles (trackers) with a diameter of 10 μm designed to follow the flow dynamics and illuminating the trackers with a laser sheet produced by Litron Nano–S–65 Nd:YAG laser source. Accordingly, the velocity can be estimated by using differences in tracker positions between created images.

3. Data processing and uncertainty analysis

In these experiments, the experimental heat flux q'' is calculated from electrical voltage (E) multiplied by current (I), measured at both ends of the heater, divided by area of the rod (A_r) with the unit of measurement of watts per meter squared (W/m2) as expressed by:

$$q'' = \frac{Q}{A_r} = \frac{E \times I}{A_r}; \tag{1}$$

The average heat transfer coefficient (h) with and without ultrasonic waves conditions is determined using Newton's equation as expressed by:

$$h = \frac{q''}{T_s - T_f}; \tag{2}$$

Where T_f is the water temperature and T_s is the average surface temperature obtained from surface temperatures of 2 points using the following relation:

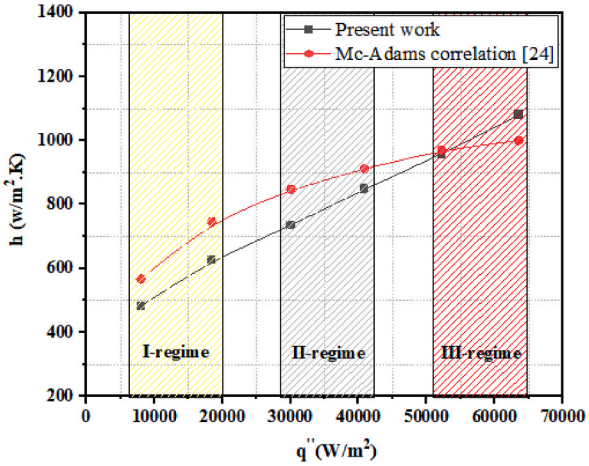


Fig. 2. Data verification of heat transfer coefficients without imposing ultrasonic waves at H = 20 mm.

$$T_{av,s} = \frac{\sum_{i=1}^{2} T_{si}}{2}; \tag{3}$$

Rayleigh number (Ra) can be calculated based on the rod length (L) and temperature difference (ΔT) as follow:

$$Ra = \frac{\Delta T \cdot g \cdot \beta \cdot L^3}{\nu . \alpha} \cdot Pr; \beta = -\frac{1}{\rho_f}\left(\frac{\partial \rho}{\partial T}\right); \tag{4}$$

To evaluate the effect of ultrasonic waves on the cooling performance, the heat transfer enhancement factor was adopted, which is defined as a ratio of average heat transfer coefficient with ultrasonic waves to that without imposing ultrasonic waves under similar conditions. Thus, heat transfer enhancement factor can be expressed as:

$$EF = \frac{h\ with \text{ ultrasonic waves}}{h\ without \text{ ultrasonic waves}}; \tag{5}$$

Due to the inaccuracies in the measurement instruments, experimental data are always subject to some level of uncertainty and, therefore, uncertainties of calculated and measured quantities were evaluated using the methodology introduced by Ref. [23]. The uncertainty of the heat transfer coefficient and heat flux were determined as follows:

$$\left(\frac{\Delta q''}{q''}\right) = \sqrt{\frac{1}{q''}\left[\left(\frac{\partial}{\partial E}(q'')\Delta E\right)^2 + \left(\frac{\partial}{\partial \mathrm{I}}(q'')\Delta \mathrm{I}\right)^2 + \left(\frac{\partial}{\partial \mathrm{A}}(q'')\Delta \mathrm{A}\right)^2\right]}; \tag{6}$$

$$\left(\frac{\Delta h}{h}\right) = \sqrt{\frac{1}{h}\left[\left(\frac{\partial}{\partial q''}(h)\Delta q''\right)^2 + \left(\frac{\partial}{\partial T_s}(h)\Delta T_s\right)^2 + \left(\frac{\partial}{\partial T_f}(h)\Delta T_f\right)^2\right]}; \tag{7}$$

The uncertainties of calculated quantities and the uncertainties of most important measured quantities are summarized in Table 1.

4. Results and discussion

During the experiments, water contained within the heat exchange vessel is preheated to a range of temperatures and its height is set at a constant of 200 mm so that the rod and ultrasonic transducer were fully submerged in water. Based on the vessel design and power supply ability of the existing electrical heater, experiments were conducted under a range of heat fluxes (q" = 8000–63521 W/m^2), range of water temperatures (T_w = 30–50 °C) along with four-rod heights, the distance between the top surface of the transducer and rod's surface, (H = 20, 35,50, 65 mm). To prove the reliability of the experimental setup and obtain reference data for comparison with subsequent results, some experiments without imposing ultrasonic waves were conducted, and the obtained results in terms of heat transfer coefficient are compared and validated with Mc-Adams's correlations found in Ref. [24] for a horizontal cylinder on the single-phase natural convection conditions as stated below.

$$Nu = 0.53\ Ra^{1/4}, for\ \ 10^4 < Ra < 10^9; \tag{8}$$

$$Nu = 0.13\ Ra^{1/3}, for\ 10^9 < Ra < 10^{12}; \tag{9}$$

The variation of heat transfer coefficients with loaded heat fluxes for the rod without imposing ultrasonic waves at (H = 20 mm) is displayed in Fig. 2. It can be observed that the heat transfer coefficient starts to increase gradually and more rapidly at high heat flux. The average deviations between the present data and Mc-Adams' correlations are within 6.7%, which could be related to the inaccuracies in the measurement instruments. In general, this accuracy is acceptable and gives reliable heat transfer results, indicating that the experimental setup and procedure used in this investigation are reliable. Based on Fig. 2, three different heat transfer regimes were assigned: the first is a natural convection regime that occurs at low heat flux; the second is a local nucleate boiling regime that occurs at medium heat flux with limited nucleate bubbles formed on the rod's surface; and the third is a subcooled nucleate boiling regime that occurs at high heat flux with more bubbles formed on the rod's surface, which explain why the present data at high heat flux conditions are higher than those predicated by Mc-Adams' correlations.

Fig. 3 shows the effects of imposing ultrasonic waves on heat transfer coefficient with different heat fluxes and different water

Table 1
Uncertainties of calculated and measured quantities.

Items	Uncertainty
Calculated quantity	
Heat transfer coefficient	1.6%
Heat flux	0.19%
Rod surface area	0.0002%
Measured quantity	
Temperature	±0.5°C
Voltage	0.04%
Current	0.03%

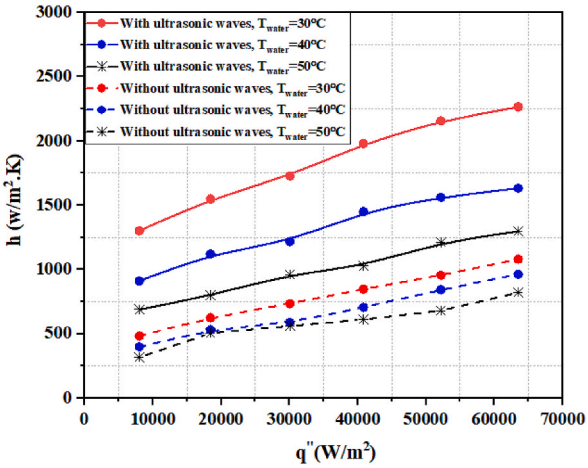


Fig. 3. Comparison of heat transfer coefficients for various heat fluxes.

temperatures but the same height (H = 20 mm). The general trend in Fig. 3 is found that the heat transfer coefficient increased with imposing ultrasonic waves. This increase is directly associated with the existence of acoustic cavitation and acoustic streaming, which results in wiping out most of the bubbles on the rod's surface in both higher and lower heat flux conditions by increasing turbulence and micro-mixing. The increasing in heat transfer coefficient is about 71.3% at water temperature (T_w = 50 °C) as compared to no ultrasonic waves. This amount of increase is acceptable when compared to the outcomes of Ref [16], which reported a (50–60%) increase in Nu using a spherical element as a heated surface at (H = 20 mm) and an ultrasonic transducer with frequency only (25 kHz). However, the increasing becomes more meaningful when the temperature of water is decreased to (T_w = 40 °C), reaching to about 96% as opposed about 132.3% for the conditions of (T_w = 30 °C). Fig. 4 shows the exemplary images of the heated surface with and without imposing ultrasonic waves. Fig. 4 (a) is taken at a lower heat flux condition (q" = 8000 W/m^2), and subfigure (b) is taken at a higher heat flux condition (q" = 52198 W/m2) without imposing ultrasonic waves, while subfigure (c) is taken at a higher heat flux condition with imposing ultrasonic waves. In Fig. 6 (a), only a few nucleate bubbles formed on the rod's surface and the natural convection heat transfer regime is dominant. While in Fig. 4 (b), unlimited nucleate bubbles are observed that can grow, move, condense, appear frequently, and the dominant heat transfer is subcooled nucleate boiling. With imposing ultrasonic waves, most of the bubbles are

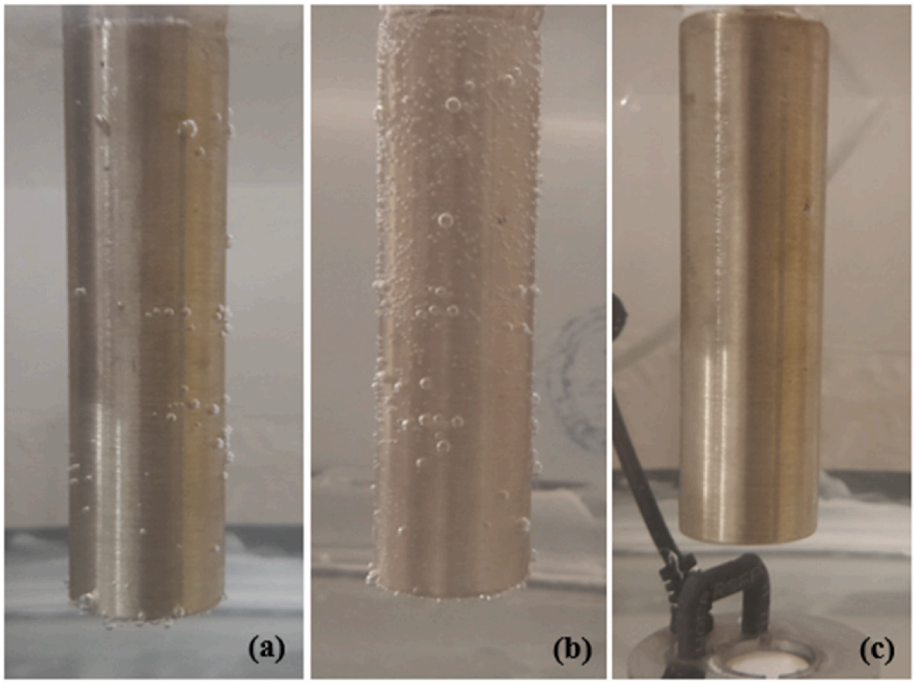


Fig. 4. Exemplary images of the heated surface during the experiments: (a) without ultrasonic waves, (q" = 8000 W/m^2); (b) without ultrasonic waves, (q" = 52198 W/m2); (c) with ultrasonic waves, (q" = 52198 W/m^2).

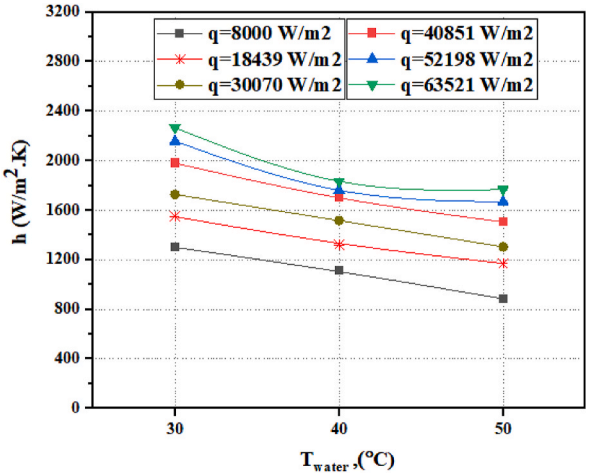


Fig. 5. Comparison of heat transfer coefficients for various water temperature.

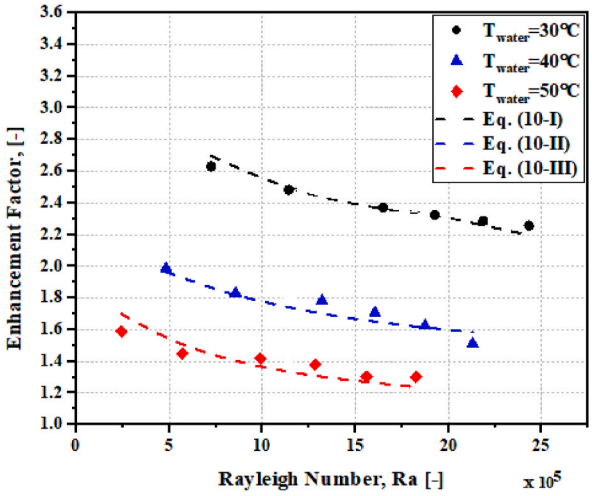


Fig. 6. Comparison of enhancement factor for various heat fluxes.

lifted off rapidly as a result of acoustic cavitation and acoustic streaming as shown in Fig. 4 (c), which create disturbances in thermal boundary layer and ultimately enhance the heat transfer rate.

Comparison of heat transfer coefficients for various water temperature is illustrated in Fig. 5. Increasing the water temperature yields a low heat transfer coefficient with a similar trend pattern in comparison with those of ($T_w = 30$ °C). The decreasing in the heat transfer coefficient with ($T_w = 40$ °C and 50 °C) are about 17.2% and about 46.8% comparative with ($T_w = 30$ °C) at the minimum heat flux of q" = 8000 W/m^2. The heat transfer enhancement factor (EF) calculated by Eq. (5), showing the practical benefit of imposing ultrasonic waves, in which heat transfer coefficient with and without imposing ultrasonic waves are evaluated simultaneously under the same operating conditions. Fig. 6 depicts the variation of heat transfer enhancement factor (EF) with Rayleigh number (Ra). Obviously, the enhancement factor tends to decrease with increasing Rayleigh number (Ra) for all cases of water temperature. The maximum enhancement factor of about 2.7 (i.e., 270%) is achieved for the operating conditions with ($T_w = 30$ °C) and (q" = 8000 W/m^2), and is higher than those with ($T_w = 40$ °C) by about 35%, while higher than other operating conditions with $T_w = 50$ °C by about 86%. Overall, it can be concluded that the ultrasonic waves with water conditions ($T_w = 30$ °C) and lower heat flux is the optimal option in terms of heat transfer enhancement, where it is rated first for all the measured parameters. The obtained results under test conditions were employed to develop the empirical correlations for further practical use as follows:

$$EF = C_1\, Ra^{C_2}, \tag{10}$$

where

I. $C_1 = 14.6$, $C_2 = -0.127$ $for \Delta T_{subcooled} = 30°C$;
II. $C_1 = 13.1$, $C_2 = -0.145$ $for \Delta T_{subcooled} = 40°C$;

III.$C_1 = 11.9, C_2 = -0.157 \quad for \Delta T_{subcooled} = 50°C$;

The above empirical correlations with two proposed empirical coefficients (C1 and C2) were expressed with the dash lines in Fig. 6. In order to confirm the accuracy of correlations, comparisons were made between predicted data from the proposed correlations and those obtained from the present work. It was found that the EF agrees well with the experimental results with average deviation of $\pm$3%.

The obtained results indicate that the effect of imposing ultrasonic waves on heat transfer enhancement is strongly depends on the waves form propagating in the water. Thus, to clarify the reason behind this significant enhancement, the velocity vectors and flow pattern of water are analyzed in the presence of an ultrasonic waves using (PIV) system. The velocity vectors and flow visualization of the acoustic streaming with imposing ultrasonic waves are shown in Fig. 7(a). Strong acoustic streams are observed emanating from ultrasonic transducer, these streams are located in the region above the ultrasonic transducer in the axial direction with a velocity ranging from 0.1 to 0.3 m/s. The presence of such acoustic streams helps in increasing water circulation inside the vessel, increasing turbulence intensity, increasing water mixing, and establishing a uniform water temperature inside the vessel, resulting in a higher heat transfer enhancement factor at certain positions. Based on Nomura et al. [25], the acoustic streams and pressure distribution in a water vessel with imposing ultrasonic waves vary with water conditions and height (distance). According to these considerations, the distance between the top surface of the ultrasonic transducer and heated surface is tested and plotted along with predicted sine wave as shown in Fig. 7(b) with respect to their possible amplitude of the ultrasonic waves. In the current water vessel, the length of ultrasonic waves is estimated to be around 55 mm. From Fig. 7(b), it can be clearly seen that the test heights (H = 20, 35, and 50 mm) are located at wavy zone and H = 65 mm located at the node zone. This indicates that the enhancement of heat transfer caused by acoustic streams relatively higher in wavy zone due to relative high pressure distribution, while it's relatively lower in node zone due to relative low pressure distribution.

Comparison of heat transfer coefficients for various tested heights under different heat fluxes but the same water temperature (T_w = 30 °C) is illustrated in Fig. 8. It can be observed that the presence of ultrasonic waves causes an increase in heat transfer coefficients, and it tends to decrease as height increases. As mentioned previously, at low heat flux (q" = 8000 W/m^2), the dominant heat transfer regime is a single-phase natural convection regime, and hence the heat transfer mechanism may depend on the interaction between water and rod surface. This reflects that the domination of acoustic streaming on the heat transfer process, which is decreases with increasing height. Conversely, at high heat flux (q" = 63521 W/m2), the mechanism of heat transfer depends on not only the interaction between water and rod surface created by acoustic streaming but also on the subcooled nucleate boiling heat transfer regime, which leads to forming more bubbles on the rod's surface. These bubbles move erratically around the rod to mix the surrounding water, bring more cool water to the rod, thereby enhance the rate of heat transfer.

For more analyzing and to facilitate the comparison of the results under the various heat flux conditions, EF plotted against heights as exhibited in Fig. 9. The enhancement factor obtained at low and heat flux showed a decreasing trend against height and the maximum values can be observed at the height of H = 20 mm. For the results obtained under medium heat flux, the trends show slightly increase and then decrease with increasing height. Besides, the results obtained under high heat flux conditions, showed an

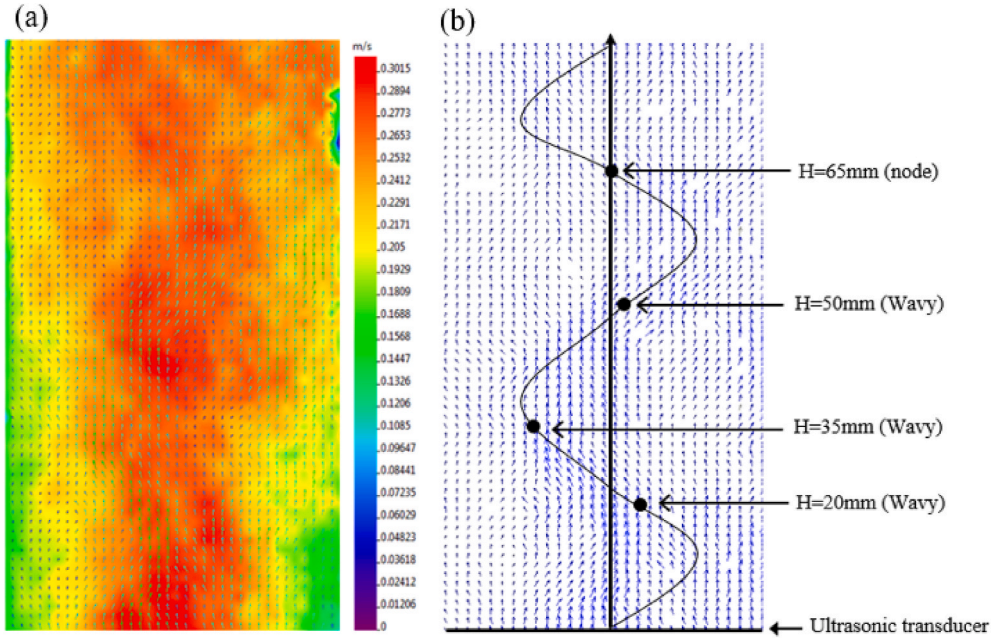


Fig. 7. (a) Velocity vector field with imposing ultrasonic waves (contour map depicts the velocity of flow); (b) Predicted amplitudes of ultrasonic waves and test heights.

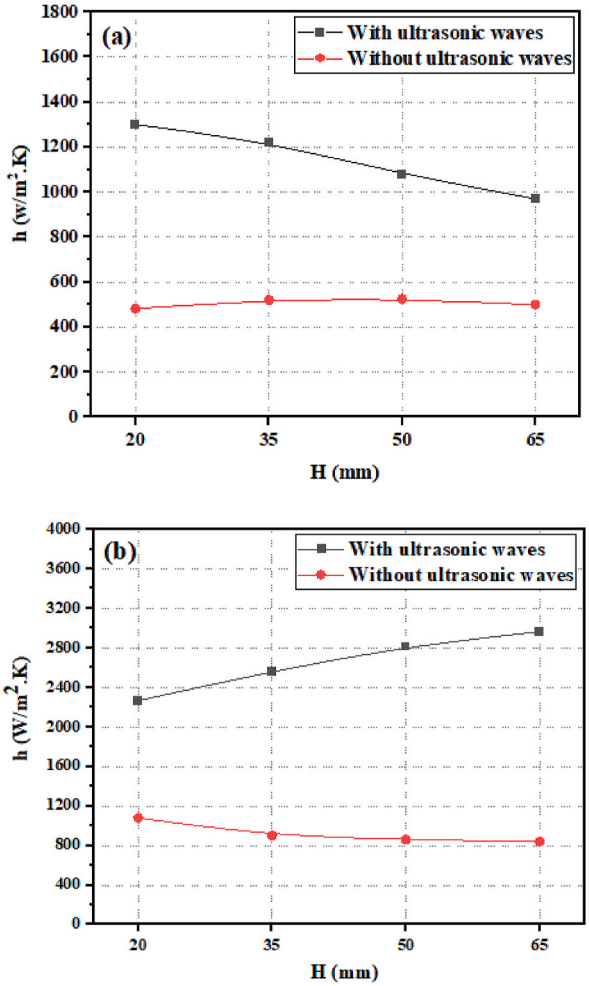


Fig. 8. Comparison of heat transfer coefficients for various tested heights with $T_w = 30$ °C under: (a) (q" = 8000 W/m^2), (b) q"=(63521 W/m^2).

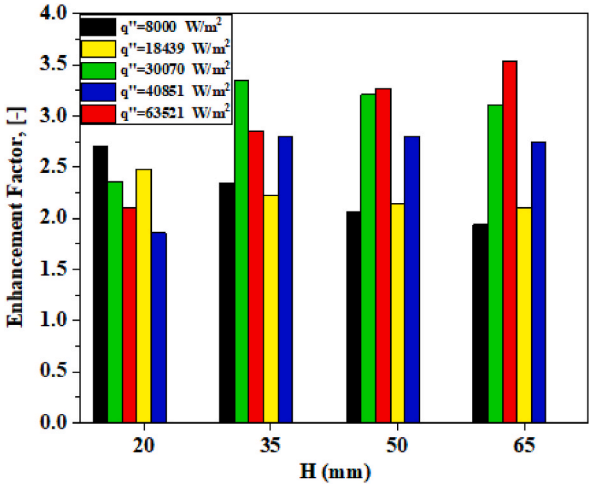


Fig. 9. Comparison of average enhancement factor for various tested heights with ($T_w = 30$ °C).

increasing trend against height and also showed higher EF values than those of medium heat flux conditions. The maximum EF of about 3.52 is achieved for the operating conditions with H = 65 mm and (q'' = 63521W/m^2), and is higher than those under (q'' = 8000 W/m^2) by about 82%, while higher than the rest heat fluxes (q'' = 18439, 30070, 40581W/m^2) by about 8, 59 and 42%, respectively.

5. Conclusion

The main objectives of the present investigation are to examine the effect of ultrasonic waves on the enhancement of heat transfer, determine the parameters that maximize the heat transfer coefficient, and obtain a systematic database under ultrasonic waves with a high frequency of 1.7 MHz. Experiments are conducted under a range of heat transfer regimes of single-phase natural convection regime, local nucleate boiling regime, and subcooled nucleate boiling regime. The important and essential conclusions can be summarized as follow:

1. The experimental results clearly proved the ability of ultrasonic waves to enhance the rate of heat transfer as well as the effect of heat flux, height and water temperature.
2. Using ultrasonic waves led to an increase in the heat transfer coefficient, the increase in the heat transfer coefficient reached its maximum value of about 133% compared to that without ultrasonic waves.
3. Increasing the water temperature led to decreasing the heat transfer coefficient. Increasing the water temperature to 40 and 50 °C has a negative effect on the heat transfer coefficient. At such temperatures, the average heat transfer coefficient decreased about 17.2% and 46.8%, respectively.
4. The heat transfer enhancement factor under low heat fluxes, i.e. single-phase natural convection regime, slightly decrease with increasing heights. However, under high heat fluxes, i.e. in the subcooled nucleate boiling regime, the heat transfer enhancement factor increased with increasing heights. This increase is the most under H = 65 mm compared to H = 20 mm at the same operating conditions.

Authorship statement

Conception and design of study: Akram H. A., Waleed Y·S.,Acquisition of data: Akram H. A., Analysis and/or interpretation of data: Akram H. A., Abduljabbar M. A., Waleed Y·S., Drafting the manuscript: _ Akram H. A., Waleed Y·S. Revising the manuscript critically for important intellectual content: _ Akram H. A., Abduljabbar M. A.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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NOMENCLATURE

I: Current, [A]
E: Voltage, [V]
Q: Heat load, [W]
A_r: Area, [m^2]
q": Heat flux, [W/m^2]
h: Heat transfer coefficient, [W/m^2. K]
T_s: surface temperature, [C]
T_w: Water temperature, [C]
$T_{ave.}$: Average surface temperature, [C]
L: length, [m]
Ra: Rayleigh number, [−]
β: Thermal expansion coefficient, [1/K]
ν: Kinematic viscosity, [m^2/s]
g: Gravitational acceleration constant, [m/s^2]
ρ: Density, [kg/m^3]
α: Thermal diffusivity, [m^2/s]